Problem Statement #1: AI-Based Human Behaviour Monitoring Using Pose Estimation and Gesture Analysis

While conducting an examination, continuous manual observation is difficult, leading to situations where suspicious behaviours may go unnoticed or be detected too late. The challenge is to develop an AI-powered video analytics system capable of monitoring seated students in real time using pose estimation and gesture analysis techniques. The system should analyse body posture, head orientation, and hand movements from surveillance camera feeds to identify behaviours that may indicate potential examination misconduct, such as repeated sideward glances, excessive head turning, body rotation toward neighbouring students, unusual hand movements, or attempts to communicate non-verbally. Unlike traditional surveillance systems that rely heavily on facial recognition or manual review of recorded footage, the proposed solution should focus on skeletal keypoints and motion patterns to infer behavioral anomalies while preserving student privacy. The system must operate in real time, track multiple students simultaneously, and generate explainable alerts that highlight specific observed behaviors rather than making direct accusations of cheating.

The system should be robust to variations in seating arrangements, camera angles, lighting conditions, and student demographics while maintaining high accuracy and minimizing false alarms.

Objectives

- Detect and track multiple seated students from live video feeds.
- Extract human pose landmarks and movement patterns.
- Identify suspicious head, body, and hand movements associated with potential examination misconduct.
- Identify objects like Mobile Phone, Paper/Chit or any unwanted objects as may be feasible
- Generate real-time alerts with interpretable explanations.
- Provide behavioral analytics and event logs for invigilator review.
- Ensure privacy-conscious monitoring by avoiding identity-based recognition wherever possible.

Expected Outcome A scalable AI-driven examination monitoring system that assists invigilators in maintaining examination integrity by automatically identifying and reporting unusual behavioral patterns in examination halls through pose estimation and gesture detection.

Risk Factor	Description	Impact
False Positives	Normal student actions (thinking, stretching, looking around) may be incorrectly flagged	Reduced trust in the system and unnecessary invigilator intervention.
False Negatives	Actual suspicious behavior may not be detected.	Potential examination malpractice goes unnoticed.
Occlusion	Students blocking each other from the camera view.	Incomplete pose estimation and tracking errors.
Poor Lighting Conditions	Low light, shadows, or glare affecting video quality.	Reduced detection accuracy.
Camera Blind Spots	Areas not covered by surveillance cameras.	Students in blind spots cannot be monitored effectively.
Head Pose Estimation Errors	Difficulty accurately detecting viewing direction from distant cameras.	Incorrect behaviour classification.
Multi-Person Tracking Failure	Identity switching or loss of student tracking across frames.	Incorrect attribution of suspicious activities.
Network Failure	Loss of connectivity between cameras and processing servers.	Interrupted monitoring and missed events.
Hardware Failure	Camera, GPU, storage, or server malfunction during examinations.	System downtime and loss of monitoring capability.
Privacy Concerns	Student concerns regarding surveillance and data collection.	Regulatory, ethical, and acceptance challenges.

Problem Statement #2: Offline Video Segmentation and ROI Detection Using Motion Estimation

Examination authorities often need to review large volumes of recorded surveillance footage after examinations to investigate incidents, verify suspicious activities, and analyze student behavior. Manual review of hours of video footage is time-consuming, labor-intensive, and prone to human error, making it difficult to efficiently identify relevant events.

The challenge is to develop an AI-powered offline video analytics system that automatically segments recorded examination hall videos and identifies Regions of Interest (ROI) using motion estimation techniques. The system should analyze temporal motion patterns within video recordings to detect areas where significant activity occurs and segment the footage into meaningful events for further review.

Unlike traditional video review methods that require continuous manual observation, the proposed solution should automatically identify motion-based regions, filter out static scenes, highlight periods of unusual activity, and generate summarized outputs for investigators and examination authorities. The system should operate on previously recorded videos, enabling efficient post-examination analysis without requiring real-time processing.

The solution should provide event-based video segmentation, ROI visualization, motion heatmaps, activity timelines, and searchable event logs to reduce the effort required for manual footage examination.

Objectives

- Process recorded examination hall videos offline.
- Detect motion regions using frame-to-frame motion estimation techniques.
- Automatically identify Regions of Interest (ROI) containing significant activity.
- Identify external objects like Mobile Phone, Paper/Chits or another other prohibited object as may be feasible
- Segment long video recordings into activity-based clips.
- Generate motion heatmaps and activity timelines.
- Reduce manual video review effort by highlighting relevant events.
- Provide event summaries and visual analytics for investigators.

Expected Outcome

A scalable offline video analytics system capable of automatically segmenting examination hall recordings and identifying motion-based Regions of Interest, enabling faster and more efficient post-examination investigations.

Challenges for Real-World Deployment

Challenge	Description
Crowded Examination Halls	Simultaneous movements from multiple students can create overlapping motion regions.
Subtle Motion Detection	Small movements such as head turns, hand gestures, or object exchanges may be difficult to distinguish from normal activity.
Camera Noise	Video compression artifacts, sensor noise, and recording distortions may generate false motion detections.
Lighting Variations	Changes in illumination, shadows, and reflections can affect motion estimation accuracy.
Camera Vibration	Slight camera movement can produce false motion across the entire frame.
Long Video Duration	Examination recordings may span several hours, requiring efficient processing and storage management.
ROI Boundary Accuracy	Accurately localizing the area responsible for motion can be challenging in dense environments.
Background Motion	Fans, curtains, lighting fluctuations, and environmental changes may create unwanted motion signals.
Event Segmentation Quality	Determining the start and end of meaningful events requires robust temporal analysis.
Multi-Camera Synchronization	Combining ROI information from multiple camera feeds may require timestamp alignment.
Scalability	Processing large numbers of recorded videos across multiple examination centres can be computationally intensive.
False Event Generation	Minor or irrelevant movements may be incorrectly classified as important events.